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Department of Mathematics
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Math 208 Major Exam I Term 253

Original exam date: 2 July 2026. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. If the solution space of the system

$$\begin{aligned} x_1 - 2x_2 + x_3 + x_4 &= 0 \\ x_1 - 2x_2 + 2x_3 - x_4 &= 0 \end{aligned}$$

is the set of all linear combinations of the form $s\mathbf{u} + t\mathbf{v}$, where s and t are real numbers and

$$\mathbf{u} = (a, b, 0, 0), \qquad \mathbf{v} = (m, n, 2, l),$$

then $\dfrac{a}{b} + \dfrac{2}{l} =$

(a) 1

(b) 2

(c) 0

(d) 4

(e) 3

2. If $y = y(x)$ is the solution of the initial value problem

$$\left(\frac{2 + \sin x}{y + 1}\right) \frac{dy}{dx} = -\cos x, \qquad y(0) = 1,$$

then $y\left(\dfrac{\pi}{2}\right)$ equals

(a) $-\dfrac{1}{3}$

(b) $-\dfrac{2}{3}$

(c) $\dfrac{4}{3}$

(d) $\dfrac{2}{3}$

(e) $\dfrac{1}{3}$

3. The general solution of the differential equation

$$xy'' + y' = 4x, \qquad x > 0$$

is given by

(a) $y(x) = 2x^2 + Ax \ln x + B$

(b) $y(x) = 4x^2 + Ax \ln x + B$

(c) $y(x) = x^2 + A \ln x + B$

(d) $y(x) = 2x^2 + A \ln x + B$

(e) $y(x) = 3x^2 + A \ln x + B$

4. The solution of the linear differential equation

$$\frac{dy}{dx} = \frac{y}{2y \ln y + y - x}$$

is given by

(a) $xy = c + x^2 \ln y$

(b) $xy = c + y^2 \ln y$

(c) $xy = cx + y^2 \ln y$

(d) $2xy = c + y^2 \ln y$

(e) $2xy = cx + y^2 \ln y$

5. If the vectors $\mathbf{u} = (1, k, 2)$, $\mathbf{v} = (2, 2, 0)$, and $\mathbf{w} = (1, 2, -2)$ in $\mathbb{R}^3$ are linearly dependent, then $k =$

(a) 1

(b) 0

(c) -2

(d) 2

(e) -1

6. A general solution of the exact differential equation

$$\left(\frac{2x}{y} - \frac{3y^2}{x^4}\right)dx + \left(\frac{2y}{x^3} - \frac{x^2}{y^2} + \frac{1}{\sqrt{y}}\right)dy = 0$$

is

(a) $\dfrac{x^2}{y} - \dfrac{y^2}{x^2} + 2\sqrt{y} = c$

(b) $\dfrac{x^2}{y^2} + \dfrac{y^2}{x^3} + \sqrt{y} = c$

(c) $\dfrac{x^2}{y} - \dfrac{y}{x^3} + \sqrt{y} = c$

(d) $\dfrac{x^2}{y} + \dfrac{y^2}{x^3} + 2\sqrt{y} = c$

(e) $\dfrac{x}{y} + \dfrac{y^2}{x^3} + 2\sqrt{y} = c$

7. If a particle is moving in a straight line with an acceleration $a(t) = \dfrac{1}{\sqrt{1-t^2}}$, initial position $x(0) = 1$, and an initial velocity $v(0) = 0$, then the position of the particle at any time t is given by

(a) $x(t) = t\sin^{-1} t + \sqrt{1-t^2}$

(b) $x(t) = 2t\sin^{-1} t + 4\sqrt{1-t^2}$

(c) $x(t) = t\sin^{-1} t + 2\sqrt{1-t^2}$

(d) $x(t) = 2t\sin^{-1} t + \sqrt{1-t^2}$

(e) $x(t) = t\sin^{-1} t + 3\sqrt{1-t^2}$

8. A thermometer with an initial reading of 10° is brought into a room maintained at a constant temperature of 30°. If the thermometer reads 20° after 2 minutes, what is the total time required for the thermometer to reach 25°?

(a) 6 minutes

(b) 5 minutes

(c) 4 minutes

(d) 3 minutes

(e) 8 minutes

9. Which one of the following subsets is a subspace of $\mathbb{R}^4$?

(a) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1 x_3 = 0$

(b) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1 - 2x_2 = 0$ and $x_3 + 5x_4 = 0$

(c) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1 \geq 0$

(d) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1 + x_2 - x_3 + 2x_4 = 1$

(e) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1^2 + x_2^2 = 1$

10. If the function $g(x)$ with $g(0) = 1$ makes the differential equation

$$[g(x) - x\cos(xy)]\,dy = \left[e^{2x} + y\cos(xy)\right]dx$$

an exact differential equation, then $g(x) =$

(a) 1

(b) $\cos(2x)$

(c) $\sin(x) + 1$

(d) $\cos(x)$

(e) e^{2x}

11. By using an appropriate substitution, the differential equation

$$\frac{dy}{dx} = 1 + \sqrt{y - 4x + 1}$$

transforms into the separable differential equation

(a) $\dfrac{du}{\sqrt{u} - 3} = dx$

(b) $\dfrac{du}{\sqrt{u} - 1} = 2\,dx$

(c) $\dfrac{du}{\sqrt{u} + 3} = dx$

(d) $\dfrac{du}{\sqrt{u} + 2} = 3\,dx$

(e) $\dfrac{du}{\sqrt{u} + 1} = dx$

12. The sum of all values of m for which $y = x^m$ is a solution of the differential equation

$$x^2y'' - 7xy' + 15y = 0$$

is

(a) 2

(b) -8

(c) 8

(d) 0

(e) -2

13. The general solution of the differential equation

$$\frac{dy}{dx} + y\cot x = \sin x, \qquad 0 < x < \pi$$

is

(a) $2y\sin x = x - \sin(2x) + c$

(b) $3y\sin x = 4x - \sin(2x) + c$

(c) $4y\sin x = x - 2\sin(2x) + c$

(d) $2y\sin x = 2x + \sin(2x) + c$

(e) $4y\sin x = 2x - \sin(2x) + c$

14. The solution of the initial-value problem

$$x\frac{dy}{dx} - y = y^2\ln x, \qquad y(1) = \frac{1}{2}$$

is given by

(a) $\dfrac{1}{y} = 1 - 2\ln x + \dfrac{1}{x}$

(b) $\dfrac{1}{y} = 1 - \ln x + \dfrac{1}{x}$

(c) $y = 1 + 2\ln x - \dfrac{1}{2x}$

(d) $y = 1 - \ln x + \dfrac{1}{x}$

(e) $\dfrac{1}{y} = 1 + 2\ln x + \dfrac{1}{x}$

15. The general solution of the differential equation

$$(x^2 + 3xy + y^2)\,dx - x^2\,dy = 0$$

is given by

(a) $\ln|x| + \dfrac{y}{x} + 1 = c$

(b) $\ln|x| + \left(1 + \dfrac{x}{y}\right)^{-1} = c$

(c) $\ln|x| + \left(1 + \dfrac{y}{x}\right)^{-1} = c$

(d) $y = x\ln|x| + cx$

(e) $y = 2x\ln|x| + cx$

Answer Key

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Q	Ans	Brief explanation
1	**(d)**	$x_3 = 2x_4$, $x_1 = 2x_2 - 3x_4$: $\mathbf{u} \parallel (2,1,0,0)$, $l = 1$, so $2+2=4$.
2	**(e)**	Separable: $y+1 = \frac{4}{2+\sin x}$, so $y(\frac{\pi}{2}) = \frac{4}{3} - 1 = \frac{1}{3}$.
3	**(c)**	$p = y'$: $(xp)' = 4x$, $p = 2x + \frac{A}{x}$, so $y = x^2 + A\ln x + B$.
4	**(b)**	Linear in x: $\frac{dx}{dy} + \frac{x}{y} = 2\ln y + 1$, $\rho = y$, so $xy = y^2 \ln y + c$.
5	**(b)**	$\det \begin{bmatrix} 1 & k & 2 \\ 2 & 2 & 0 \\ 1 & 2 & -2 \end{bmatrix} = 4k = 0$, so $k = 0$.
6	**(d)**	$F = \int M\,dx = \frac{x^2}{y} + \frac{y^2}{x^3} + g(y)$, $g' = \frac{1}{\sqrt{y}}$, so $g = 2\sqrt{y}$.
7	**(a)**	$v = \sin^{-1} t$; by parts $x = t\sin^{-1} t + \sqrt{1-t^2} + C_2$ with $C_2 = 0$.
8	**(c)**	$T = 30 - 20e^{-kt}$, $e^{-2k} = \frac{1}{2}$; $e^{-kt} = \frac{1}{4} = e^{-4k}$, so $t = 4$ min.
9	**(b)**	Only the homogeneous linear system $x_1 - 2x_2 = 0$, $x_3 + 5x_4 = 0$ gives a subspace.
10	**(a)**	$M = e^{2x} + y\cos xy$, $N = -g + x\cos xy$: $M_y = N_x \Rightarrow g' = 0$, so $g = 1$.
11	**(a)**	$u = y - 4x + 1$: $u' = y' - 4 = \sqrt{u} - 3$, so $\frac{du}{\sqrt{u}-3} = dx$.
12	**(c)**	$m(m-1) - 7m + 15 = m^2 - 8m + 15 = (m-3)(m-5)$: $m = 3, 5$, sum 8.
13	**(e)**	$\rho = \sin x$: $(y\sin x)' = \sin^2 x$, so $y\sin x = \frac{x}{2} - \frac{\sin 2x}{4} + C$.
14	**(b)**	Bernoulli $n = 2$, $v = \frac{1}{y}$: $(xv)' = -\ln x$, $v = 1 - \ln x + \frac{1}{x}$ (from $v(1) = 2$).
15	**(c)**	Homogeneous, $v = y/x$: $xv' = (v+1)^2$, so $\ln\lvert x\rvert + (1+\frac{y}{x})^{-1} = c$.